CP05 ARTIFICIAL INTELLIGENCE & EMERGING TECHNOLOGY

Prompt Engineering Cheat Sheet

Panduan Beginner Untuk Tulis Prompt Yang Lebih Baik

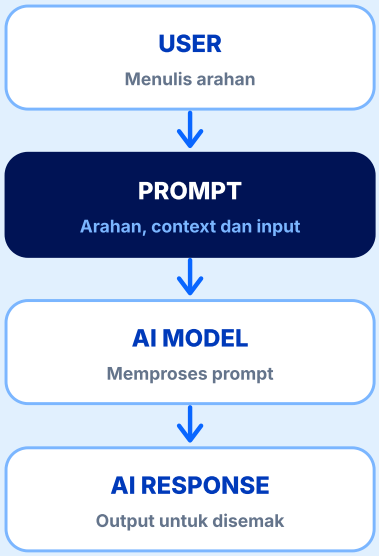
Better instructions. Clearer context. More useful AI responses.

Prompt engineering ialah proses mereka bentuk dan memperbaiki arahan yang diberikan kepada AI model untuk mendapatkan output yang lebih relevan dengan keperluan kita.

Ia cara kita menyusun arahan supaya AI lebih memahami lima perkara di bawah.

From You To AI Response

Aliran asas satu prompt.



1. **Task** Apa yang perlu dilakukan.
2. **Tujuan** Kenapa task tersebut perlu dilakukan.
3. **Input** Maklumat apa yang perlu digunakan.
4. **Constraints** Apakah batasan yang perlu dipatuhi.
5. **Format** Bagaimana output perlu disampaikan.

Instruction, Context dan Input

Tiga perkara ini sering bercampur.

INSTRUCTION
Apa yang AI perlu buat, termasuk task, batasan dan format output.

CONTEXT
Latar belakang situasi, audience dan tahap pengetahuan kita.

INPUT
Bahan sebenar yang AI perlu guna, seperti teks, data atau code.

Prompt Boleh Terdiri Daripada

Role / Context + Task + Input + Constraints + Output format + Examples

INGAT
Output AI masih perlu disemak. Prompt yang baik tidak menjamin semua jawapan tepat.

AI tidak semestinya perlukan prompt yang panjang dan complicated. Yang penting: arahan jelas, context cukup, task spesifik dan output yang dimahukan.

MAIN TAKEAWAY

6 Components Of A Structured Prompt | Definition, tujuan dan contoh.

COMPONENT 01
ROLE / CONTEXT

Berikan AI peranan atau context yang sesuai supaya respons mempunyai perspektif dan latar belakang yang relevan.

EXAMPLE
```
Act as a beginner-friendly programming tutor.
```

CONTEXT EXAMPLE
```
Saya seorang beginner yang baru mula belajar Python dan belum mempunyai pengalaman programming.
```

WHY IT MATTERS
Membantu AI memahami audience, tahap pengetahuan dan situasi pengguna.

COMPONENT 02
TASK

Nyatakan dengan jelas apa yang AI perlu lakukan.

EXAMPLE
```
Explain the difference between a list and a dictionary in Python.
```

WHY IT MATTERS
Task yang spesifik mengurangkan kekeliruan tentang hasil yang diminta.

COMPONENT 03
INPUT

Sediakan maklumat, bahan atau data yang perlu digunakan oleh AI.

EXAMPLE
```
Use the following code as the basis of your explanation: [paste code here].
```

INPUT MAY INCLUDE
Text · Documents · Data · Code · Examples · Background information

WHY IT MATTERS
AI memerlukan maklumat yang relevan untuk menghasilkan respons yang sesuai.

COMPONENT 04
CONSTRAINTS

Tetapkan batasan dan syarat yang perlu dipatuhi.

EXAMPLE
```
Maximum 300 words. Use simple language.
```

MORE EXAMPLES
Explain for a beginner · Use simple language · Maximum 300 words · Do not use advanced technical jargon · Do not invent facts · Use only the supplied information

WHY IT MATTERS
Constraints membantu menjadikan output lebih konsisten dengan keperluan pengguna.

COMPONENT 05
OUTPUT FORMAT

Nyatakan bagaimana jawapan perlu disusun atau dipersembahkan.

EXAMPLE
```
Present the answer as a comparison table.
```

FORMAT OPTIONS
Bullet points · Comparison table · Step-by-step guide · Checklist · JSON · Email draft · Summary

WHY IT MATTERS
AI boleh menghasilkan kandungan yang sama dalam format berbeza bergantung kepada keperluan pengguna.

COMPONENT 06
EXAMPLES

Berikan contoh output atau gaya yang diinginkan untuk membantu AI memahami expectation pengguna.

EXAMPLE
```
Follow this structure: Definition → Explanation → Example → Key takeaway.
```

WHY IT MATTERS
Examples can clarify the expected structure, style or level of detail. Ia membantu memandu output tetapi tidak menjamin AI meniru dengan tepat.

IMPORTANT CLARIFICATION
Examples adalah pilihan. Tidak setiap prompt perlukan kesemua enam komponen. Gunakan komponen mengikut kerumitan task.

Copy, Paste & Customise | Template boleh guna semula.

MASTER PROMPT TEMPLATE

```
ROLE / CONTEXT:
Act as [role or expertise].

My background/context:
[Explain your situation, audience or objective.]

TASK:
I want you to [describe the specific task].

INPUT:
Use the following information:
[Paste text, data, code or relevant material.]

CONSTRAINTS:
- [Requirement 1]
- [Requirement 2]
- [Requirement 3]
- [Things to avoid]

OUTPUT FORMAT:
Present the response as:
[Table / Bullet Points / Step-by-Step / Checklist / Other]

EXAMPLES:
Follow this example or structure:
[Provide a reference example if available.]

ADDITIONAL INSTRUCTION:
If important information is missing, state what is missing.
Do not invent facts. Clearly distinguish assumptions from verified information.
```

INGAT
Tak perlu isi semua bahagian untuk setiap prompt. Pilih komponen yang relevan dengan task.

Ganti Bahagian Dalam Kurungan | Contoh sahaja. Sesuaikan dengan keperluan sendiri.

REPLACE THIS	WITH YOUR OWN, FOR EXAMPLE
`[role or expertise]`	a patient maths tutor
`[describe the specific task]`	create a 4-week study plan for my SQL exam
`[Paste text, data, code...]`	your lecture notes or the error message
`[Requirement 1], [Requirement 2]`	Use simple language. Maximum 300 words.
`[Table / Bullet Points / ...]`	Weekly table

From Vague Prompt To Better Prompt

Satu contoh yang sama: belajar Python.

BEFORE: VAGUE PROMPT

```
"Teach me Python."
```

KENAPA VAGUE?

- No information about the learner's current level.
- No specific learning objective.
- No preferred teaching style.
- No requested output structure.

AFTER: STRUCTURED PROMPT

STRUCTURED PROMPT

```
ROLE:
Act as a beginner-friendly Python tutor.

CONTEXT:
I am completely new to programming.

TASK:
Teach me Python variables and data types.

CONSTRAINTS:
- Use simple explanations.
- Avoid advanced technical jargon.
- Use practical everyday examples.
- Explain one concept at a time.

OUTPUT FORMAT:
1. Definition.
2. Simple explanation.
3. Python code example.
4. Expected output.
5. Short practice exercise.

EXAMPLES:
Use examples related to shopping and daily activities.

If a concept has multiple interpretations,
explain the relevant distinction.
```

What Improved?

- **01 Clear audience** — complete beginner
- **02 Specific task** — variables and data types
- **03 Relevant context** — new to programming
- **04 Defined limitations** — simple, one concept at a time
- **05 Structured output** — five-part format
- **06 Practical examples** — shopping and daily activities

IMPORTANT

A structured prompt can improve relevance and clarity, but it does not guarantee factual correctness.

5 Prompt Templates For Everyday Tasks

Learning, summarising, writing, problem-solving dan coding.

01 Learning

LEARNING

```
Act as a patient tutor.

Teach me [topic] as a complete beginner.

Explain:
- What it means.
- Why it matters.
- How it works.
- A simple real-world example.

Use simple language and finish with
3 questions to test my understanding.
```

02 Summarising

SUMMARISING

```
Summarise the following material.

Requirements:
- Identify the main ideas.
- Preserve important details.
- Use clear headings and bullet points.
- Do not add information that is not in the source.
- Highlight unclear or missing information.

Material:
[Paste text here]
```

03 Writing

WRITING

```
Act as a professional content writer.

Create a [type of content] about [topic].

Target audience:
[audience]

Tone:
[tone]

Key points:
[points to include]

Avoid:
[things to avoid]

Output:
[required format and length]
```

04 Problem-solving

PROBLEM-SOLVING

```
Help me analyse this problem.

Problem:
[describe the problem]

Context:
[provide relevant details]

Please:
1. Identify the main issue.
2. Break it into smaller parts.
3. Explain possible causes.
4. Suggest practical solutions.
5. Identify assumptions and uncertainties.

Do not assume missing information is true.
```

05 Coding

CODING

```
Act as a programming tutor.

Help me understand this code:

[paste code]

Explain:
1. What the code does.
2. How it works step by step.
3. What each important section means.
4. Potential errors or edge cases.
5. How I can improve it.

Assume I am a beginner.
Do not rewrite the entire code unless necessary.
```

IMPORTANT

Ini starting point yang boleh diguna semula. Sesuaikan arahan mengikut task, maklumat yang ada dan hasil yang dimahukan.

Before You Send Your Prompt... | Semak sebelum tekan send.

- [ ] Is my task clear and specific?
- [ ] Have I provided enough context?
- [ ] Have I included the necessary input?
- [ ] Have I stated important constraints?
- [ ] Have I specified the desired output format?
- [ ] Have I included an example if necessary?
- [ ] Have I mentioned what AI should avoid?
- [ ] Have I explained what to do if information is missing?

Don't Trust Every AI Answer | Lima amalan verification.

01 CHECK IMPORTANT FACTS
Verify names, dates, statistics and technical claims.

02 ASK FOR SOURCES
Request credible sources when factual research is involved. Then open and verify the sources independently.

03 CHECK THE ORIGINAL MATERIAL
When working with documents, compare AI summaries and claims with the original content.

04 TEST GENERATED CODE
Run and test code in an appropriate environment before using it.

05 PROTECT PERSONAL INFORMATION
Avoid pasting passwords, sensitive personal data or confidential business information into AI tools unless the service and applicable policies are appropriate for that data.

PERINGATAN

Prompt yang bagus boleh membantu AI memahami arahan. Tetapi ia tidak menjamin jawapan yang sentiasa tepat.

Better prompts start with clearer thinking. FINAL TAKEAWAY

Download template ini dan cuba ubah ikut kerja korang.
Prompt engineering membantu kejelasan dan relevansi, bukan jaminan ketepatan.

yusufaimxn
Technology, Made Understandable.